

TIGER CS TICKETING SYSTEM

Application and Solution Architecture

- Built on ASP.NET Core, .NET 10 (LTS)

- Internal Pilot Architecture

Business Objective

1

Centralize

All customer-service requests captured in a single system of record

2

Integrate

Phone (Genesys), CRM, and internal departments into one workflow

3

Track

Ownership, SLA due dates, escalation, and resolution for every ticket

4

Provide

Auditability and management visibility across the full ticket lifecycle

System Context Architecture

INTERNAL USERS

CS Agent

Department Employee / Head

Management (CS Mgr · GM · Chairman)

System Admin / Reporting User

Customer (phone)

via Genesys (phone)

APPLICATION BOUNDARY

Tiger CS Ticketing System

**Modular Monolith
(ASP.NET Core, .NET 10)**

Web UI · API · Application Modules
Domain · Infrastructure · Integrations

SQL Server

No customer login or portal exists — all customer contact is agent-mediated (phone via Genesys) or an outbound email notification.

EXTERNAL SYSTEMS

CRM

Genesys

Office 365 Email

File Storage

Application Architecture

Modular monolith — one deployable unit, clean internal module boundaries

Web UI

ASP.NET Core API

APPLICATION MODULES

Identity & Access

CRM Verification

Intake

Ticketing

SLA & Escalation

Genesys Integration

Notifications

Attachments

Audit

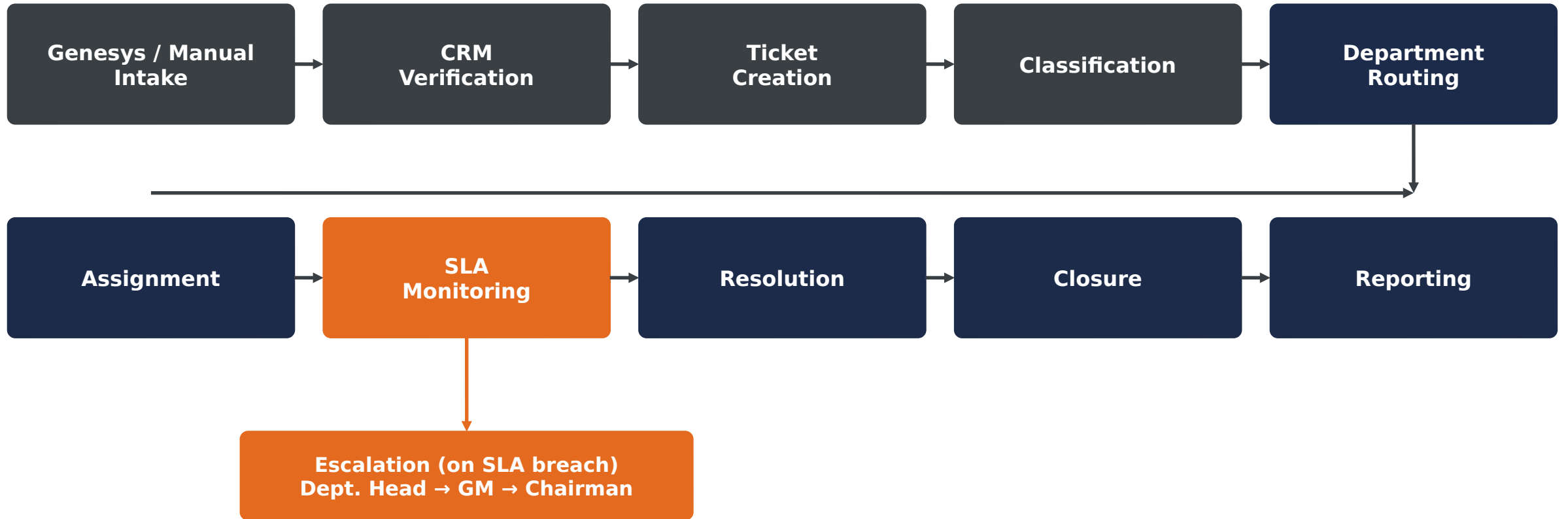
Reporting

Domain

Infrastructure | Integrations (CRM · Genesys · Email · Storage)

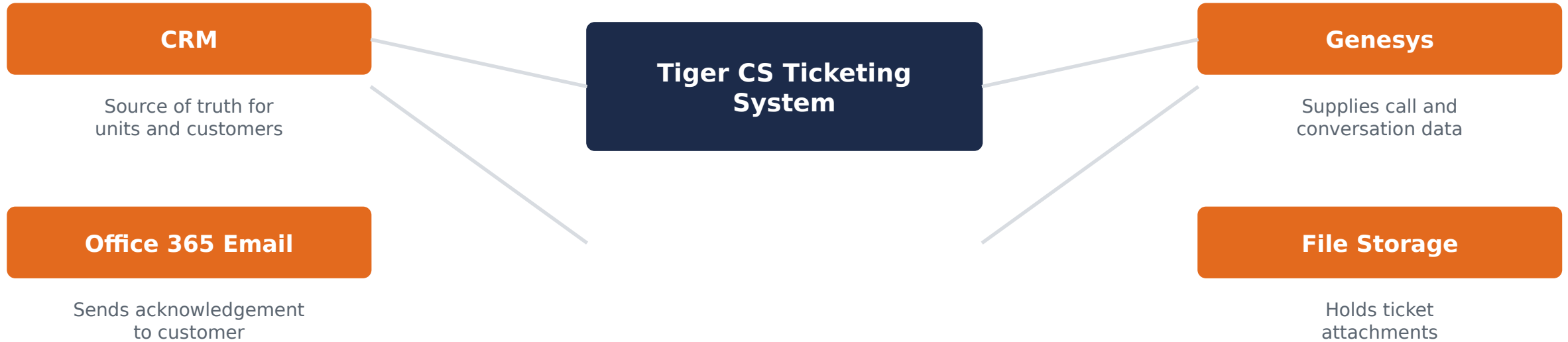
SQL Server

Main Ticket Flow



Escalation runs alongside the main flow whenever a ticket's SLA is breached, independent of ticket status.

Integration Architecture



Outbox & Idempotency

Every integration effect is written transactionally and dispatched safely — protecting against duplicate or lost processing on retries and outages.



Feature Flags

Each integration (notably Genesys) is activated behind a feature flag — allowing safe, staged rollout without a code redeploy.

Security Architecture



ASP.NET Core Identity

Staff authentication — no customer accounts exist



JWT Authentication

Issued on login, validated on every API request



Role- & Department-Based Authorization

Every action checked server-side against policy



Nine Approved Roles

From CS Agent through Chairman/CEO and System Administrator



Account Deactivation

Validated on every request, not only at login



Full Audit Trail

Append-only — no update or delete path



Secrets & Transport

Secrets kept outside source control; HTTPS everywhere

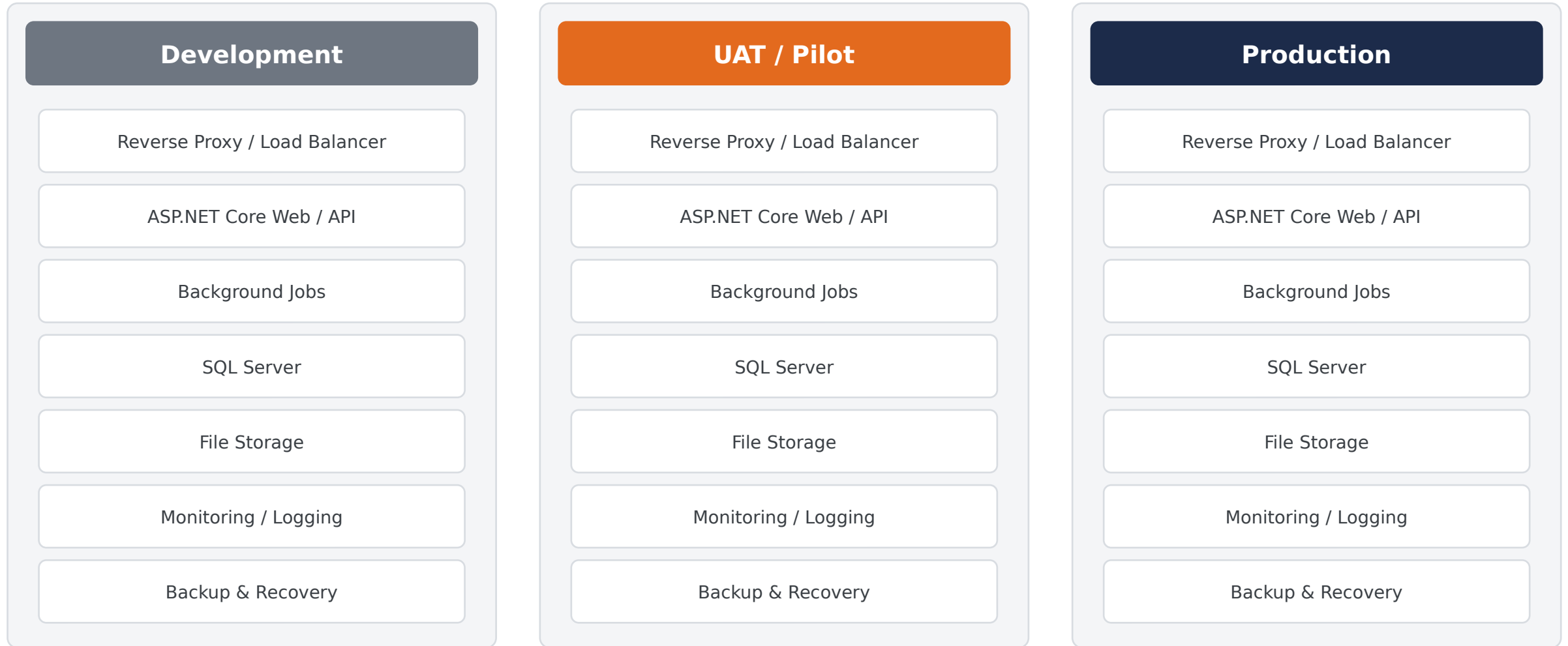


Least-Privilege Access

No production credentials ever stored in GitHub

Deployment Architecture

Target: Tiger-approved hosting environment (final hosting provider to be confirmed by Tiger IT)



Delivery Scope and Roadmap



Delivery model: Four-week, one-developer, AI-assisted pilot



Pilot (This Delivery)

A functional internal proof of the approved architecture, deployed to a non-production environment for evaluation.



Production Ready

A separate, later milestone — its own scope, hardening, and go-live approval, not implied by pilot completion.

Key Risks and Management Decisions

- ! CRM and Genesys integration contract availability
- ! Schedule pressure from a single-developer delivery model
- ! UAT participation from CS and department staff
- ! SQL Server and target environment readiness
- ! Independent security review before pilot go-live

Production go-live requires separate, explicit management approval.